



Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

035033-Intro to Sensor-Integrated Systems

Lecture V: Reliability

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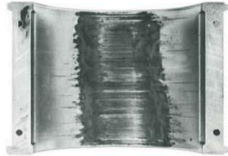
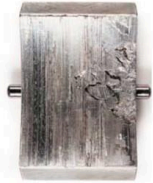
Background

- ❑ Space shuttle can take astronauts to space
- ❑ O-ring seal failure caused the explosion



Weare and tear failure of components

- ❑ The same thing happens to instruments





学院概况

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学院简介

1985年,我国国防科技界和教育界著名专家、可靠性系统工程事业的奠基人和开拓者杨为民教授,顺应国家战略重大需求,组建了北航工程系统工程系和可靠性工程研究所。2010年,工程系统工程系更名为可靠性与系统工程学院。学院和研究所实行“一个实体、两块牌子”的运行机制,历经近40年的发展,形成了以“为民精神”为核心的学院文化,秉承“开拓创新、敢为人先,需求牵引、专业推动,学科龙头、科教统筹,团队优势、集体发展”的工作理念,走过了一条“面向工程服务、开辟科研领域、创建培养体系”的特色创新之路,(学院(研究)所协同,构建了包括人才培养、科学研究、工程应用、社会服务在内的可靠性专业生态圈,成为国内可靠性领域的领军单位。

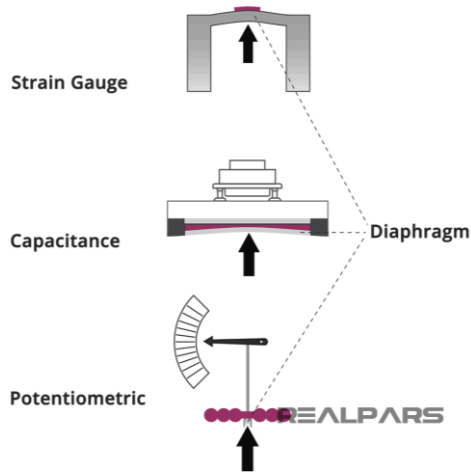
学院为北航安全科学与工程一级学科的主建单位,拥有安全科学与工程一级学科博士点、系统工程二级学科博士点、电子信息和机械专业型硕、博士学位授权点。拥有飞行器质量与可靠性和安全工程两个本科专业,其中飞行器质量与可靠性专业为国家一流本科专业建设点,安全工程专业为北京市一流本科专业建设点。现有本科生216人,研究生519人,其中硕士研究生311人,博士研究生208人。

Wear/tear and failure happens to instruments too

❑ Which one is the most reliable?



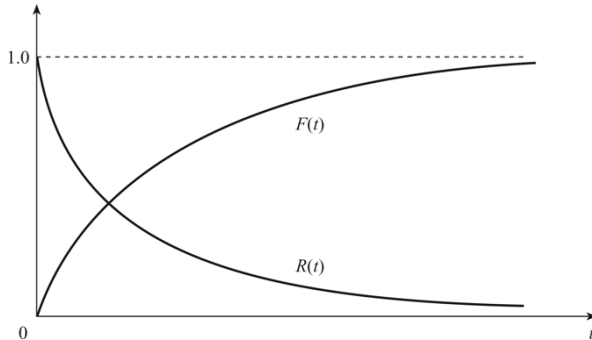
pH sensor



Strain gauge

Reliability and Unreliability

- ❑ Reliability $R(t)$ - the probability that the element or system will operate to an agreed level of performance, for a specified period, subject to specified environmental conditions
- ❑ Example: measure 100 times, 98 times success, then reliability is 98%.
- ❑ Unreliability $F(t) = 1 - R(t)$, for this example 2%.



Practical reliability definitions

❑ Mean down time: $MDT = \frac{\text{total down time}}{\text{number of failures}} = \frac{1}{N_F} \sum_{j=1}^{N_F} T_{Dj}$

❑ Total up time $NT - \sum_{j=1}^{N_F} T_{Dj} = NT - N_F * MDT$

❑ Mean Time Between Failures $MTBF = \frac{NT - N_F * MDT}{N_F} [yr]$

❑ Mean failure rate $\bar{\lambda} = \frac{1}{MTBF} = \frac{N_F}{NT - N_F * MDT} [yr^{-1}]$

❑ Availability $A = \frac{\text{total up time}}{\text{test interval}} = \frac{NT - N_F * MDT}{NT}$

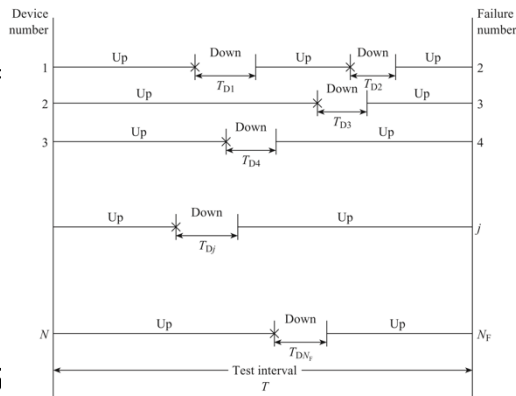
❑ Unavailability $U = 1 - A$

❑ Example:

If 150 faults are recorded for 200 transducers over 1.5 years with a mean down time of 0.002 years.

Then:

The observed MTBF is _____ years and the mean failure rate is _____ yr^{-1} , $A =$ _____.



Instantaneous failure rate $\lambda(t)$

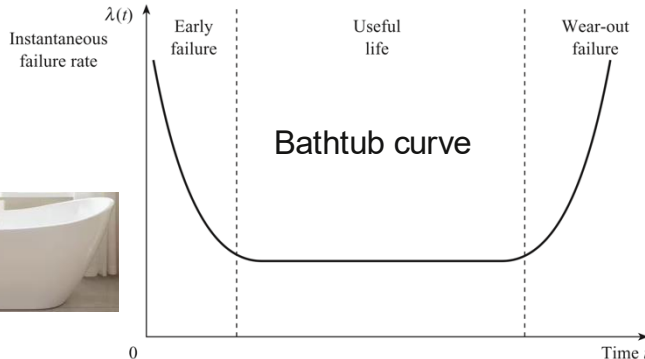
❑ Mean failure rate $\bar{\lambda} = \frac{1}{\text{MTBF}} = \frac{N_F}{NT - N_F * \text{MDT}} [\text{yr}^{-1}]$

❑ Instantaneous failure rate $\lambda(t) = \frac{N_{F \rightarrow 0}}{NT - N_F * \text{MDT} \rightarrow 0} [\text{yr}^{-1}]$



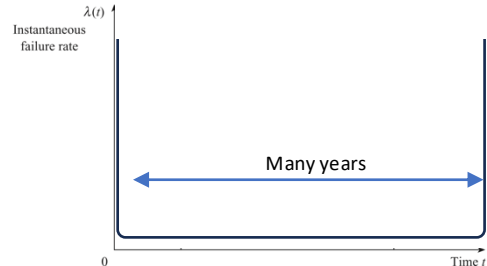
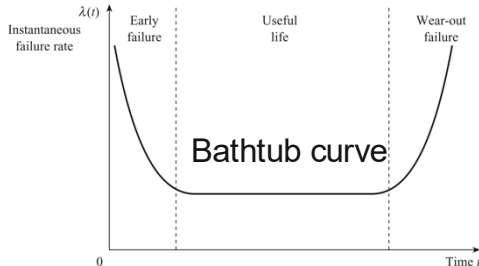
Typical forms of failure rate function

- ❑ **Early failure region:** characterized by $\lambda(t)$ decreasing with time. When items are new, especially if the element is of a new design, early failures can occur due to design faults, poor quality components, manufacturing faults, installation errors, operator and maintenance errors; the latter may be due to unfamiliarity with the product.
- ❑ The hazard rate falls as design faults are rectified, weak components are removed, and the user becomes familiar with installing, operating and maintaining the element.



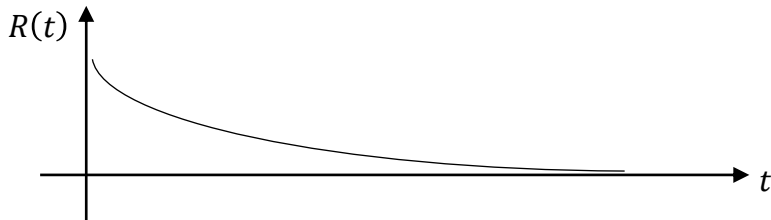
Typical forms of failure rate function

- ❑ **Useful life regionis:** useful life regionis characterised by a low, constant failure rate. Here all weak components have been removed: design, manufacture, installation, operating and maintenance errors have been rectified so that failure is due to a variety of unpredictable lower-level causes.
- ❑ **Wear-out region:** characterised by $\lambda(t)$ increasing with time as individual items approach the end of the design life for the product; long-life components which make up the element are now wearing out.
- ❑ Many measurement elements have a useful life region lasting many years, so that a constant failure rate model is often a good approximation $\lambda(t) = \lambda$.



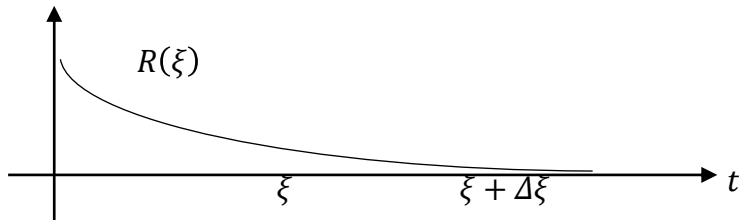
Mean failure rate $\bar{\lambda}$

- ❑ Mean failure rate $\bar{\lambda} = \frac{1}{\text{MTBF}} = \frac{N_F}{NT - N_F * \text{MDT}} [\text{yr}^{-1}] = \text{Constant}$
- ❑ $\lambda = \frac{f(t)}{R(t)}$, where $f(t) = -\frac{dR(t)}{dt}$ is probability density function of failure time
- ❑ Then $\lambda = \frac{-\frac{dR(t)}{dt}}{R(t)} \Rightarrow R(t) = e^{-\lambda t}$



Mean failure rate $\bar{\lambda}$

- ❑ Mean failure rate $\bar{\lambda} = \frac{1}{\text{MTBF}} = \frac{N_F}{NT - N_F \cdot \text{MDT}} [\text{yr}^{-1}] = \text{Constant}$
- ❑ n items of an element survive up to time $t = \xi$ and that Δn items fail during the small-time interval $\Delta \xi$ between ξ and $\xi + \Delta \xi$.
- ❑ The unconditional probability $\Delta F = R(\xi) \lambda \Delta \xi \Rightarrow \frac{\Delta F}{R(\xi)} = \lambda \Delta \xi$
- ❑ Because $F = 1 - R$, then $\Delta F = \Delta 1 - \Delta R = -\Delta R \Rightarrow -\frac{\Delta R}{R(\xi)} = \lambda \Delta \xi$
- ❑ $\lim_{\Delta R \rightarrow 0, \Delta \xi \rightarrow 0} \Rightarrow -\frac{dR}{R(\xi)} = \lambda d\xi \Rightarrow -\int_{R(0)}^{R(t)} \frac{dR}{R(\xi)} = \int_0^t \lambda d\xi \Rightarrow e^{-\int_0^t \lambda d\xi} = e^{-\lambda t}$

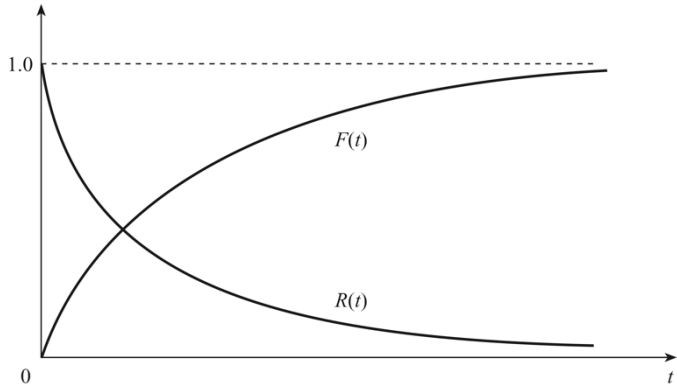
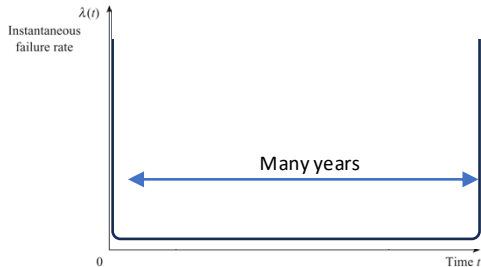


Typical forms of failure rate function

□ $R(\xi) = e^{-\int_0^t \lambda(\xi) d\xi}$

□ Reliability $R(t) = e^{-\lambda \int_0^t d\xi} = e^{-\lambda t}$

□ Unreliability $F(t) = 1 - R(t) = 1 - e^{-\lambda t}$

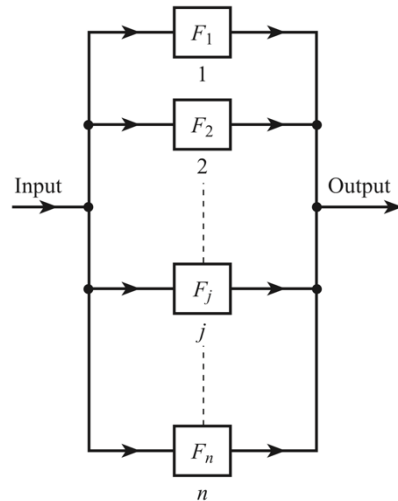
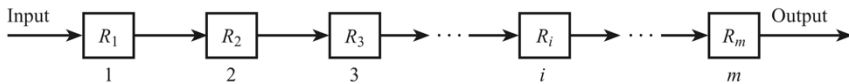


Reliability of series and parallel system

❑ Series system: $R_{SYST} = R_1 R_2 R_3 \dots R_N = e^{-\lambda_1 t} e^{-\lambda_2 t} e^{-\lambda_3 t} \dots e^{-\lambda_N t} = e^{-(\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_N)t}$

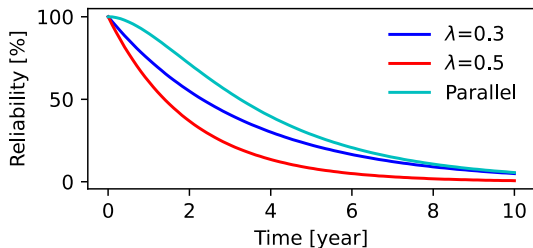
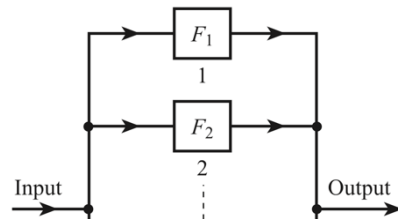
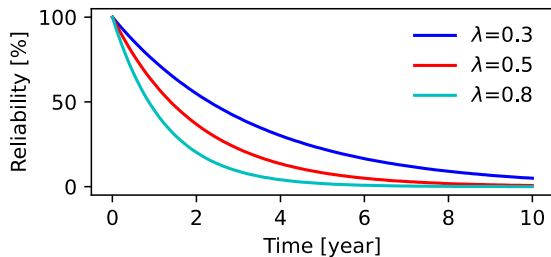
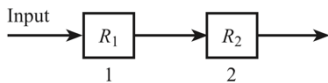
❑ Parallel system: $F_{SYST} = F_1 F_2 F_3 \dots F_N = (1 - e^{-\lambda_1 t})(1 - e^{-\lambda_2 t})(1 - e^{-\lambda_3 t}) \dots (1 - e^{-\lambda_N t})$

❑ $R_{SYST} = 1 - F_{SYST} = 1 - (1 - e^{-\lambda_1 t})(1 - e^{-\lambda_2 t})(1 - e^{-\lambda_3 t}) \dots (1 - e^{-\lambda_N t})$



Reliability of series and parallel system

- Series system: $R_{SYST} = R_1 R_2 = e^{-0.3t} e^{-0.5t} = e^{-(0.3+0.5)t}$
- Parallel system: $F_{SYST} = F_1 F_2 = (1 - e^{-0.3t})(1 - e^{-0.5t})$
- $R_{SYST} = 1 - F_{SYST} = 1 - (1 - e^{-0.3t})(1 - e^{-0.5t})$



Code for the example

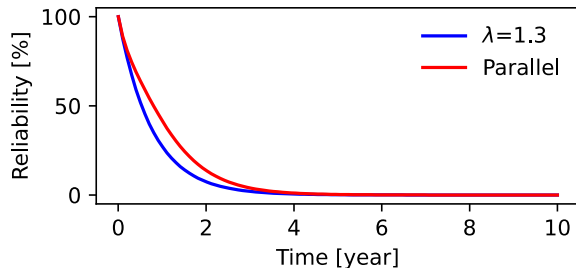
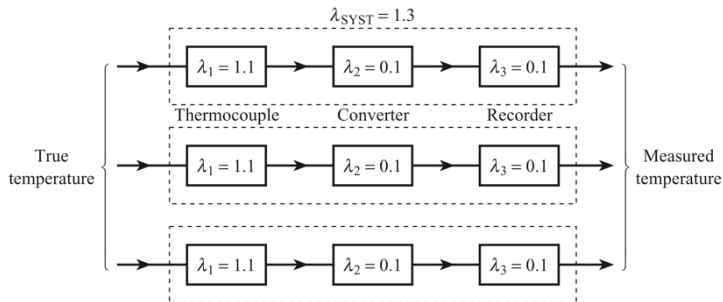
```
from numpy import *  
from matplotlib.pyplot import *
```

```
N = 100  
t = linspace(0.0001, 10, N) # time in years  
figure(figsize=[4, 2])  
plot(t, exp(-0.3*t)*100, 'b', label=r'$\lambda=0.3$')  
plot(t, exp(-0.5*t)*100, 'r', label=r'$\lambda=0.5$')  
plot(t, exp(-0.8*t)*100, 'c', label=r'$\lambda=0.8$')  
legend()  
xlabel('Time [year]')  
ylabel('Reliability [%]')  
tight_layout()  
savefig('temp.pdf')
```

```
figure(figsize=[4, 2])  
plot(t, exp(-0.3*t)*100, 'b', label=r'$\lambda=0.3$')  
plot(t, exp(-0.5*t)*100, 'r', label=r'$\lambda=0.5$')  
plot(t, 100-(1-exp(-0.3*t))*(1-exp(-0.5*t))*100, 'c',  
label='Series')  
legend()  
xlabel('Time [year]')  
ylabel('Reliability [%]')  
tight_layout()  
savefig('temp.pdf')
```

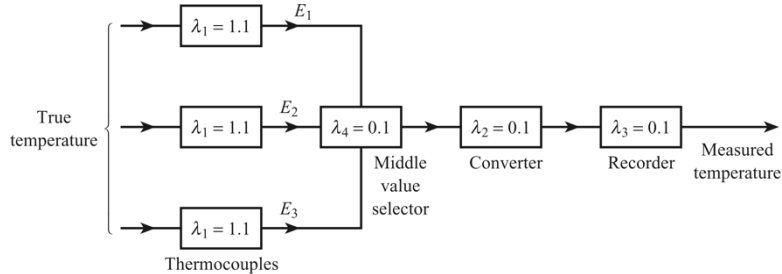
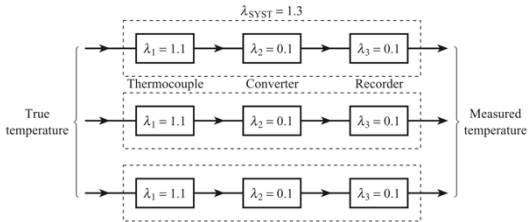
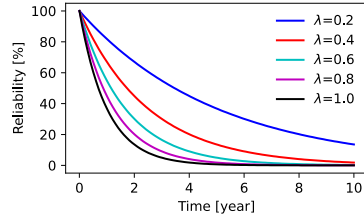

Example

- ❑ Reliability of subsystems: $R_1 = R_2 = R_3 = e^{-(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \dots)t} = e^{-1.3t}$
- ❑ Reliability of them combined: $e^{-1.3t}$
- ❑ Why? – redundancy
- ❑ At $t = 0.5$ yr, $R_1 = R_2 = R_3 = 0.52$
- ❑ Parallel $R = 1 - (1 - R_1)(1 - R_2)(1 - R_3) = 0.89$



Middle value selector

- ❑ The left parallel system, while reliable, is expensive
- ❑ **Middle value selector:**
- ❑ Example: [5.1, 5.1, 5.1], one goes wrong [5.1, 5.1, 0], [5.1, 5.1, 50], two goes wrong [5.1, 0, 50] still working
- ❑ If two goes wrong [5.1, 50, 50], [5.1, 0, 0] not working, but two goes wrong at the same time has a small chance

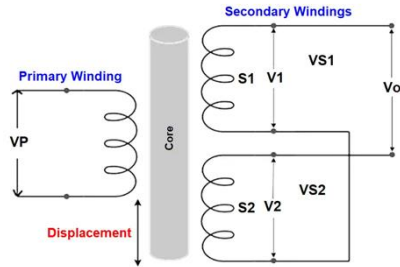


Design for reliability

- ❑ **Element selection.** Only elements with well-established failure rate data/models should be used. inductive LVDT displacement sensor is inherently more reliable than a resistive potentiometer;
- ❑ **Environment.** The environment in which the element is to be located should first be defined and the element should consist of components and elements which are capable of withstanding that environment.



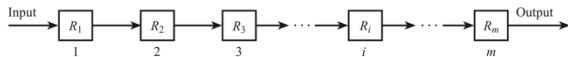
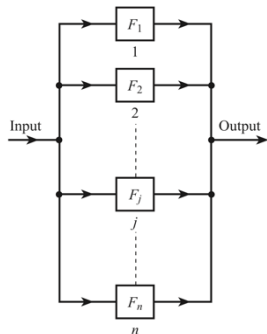
Position sensor



Position sensor

Design for reliability

- ❑ **Minimum complexity.** For a series system, the system failure rate is the sum of the individual component/element failure rates. Thus, the number of components/elements in the system should be the minimum required.
- ❑ **Redundancy.** The use of several identical elements/systems connected in parallel increases the reliability of the overall system.
- ❑ **Diversity.** In practice faults can occur which cause more than one element in each system to fail simultaneously. A given function is carried out by two systems in parallel, but each system is made up of different elements with different operating principles. Example: a temperature measurement system made of one electronic and one pneumatic in parallel.

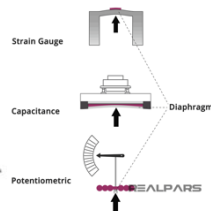


Observed average failure rates for instruments

Instrument	Environment	Experience (item-years)	No. of failures	Failure rate (failures/yr)
Chemical analyser, Oxygen	Poor, chemical/ship	4.34	30	6.92
pH meter	Poor, chemical/ship	28.08	302	10.75
Conductivity indicator	Average, industrial	7.53	18	2.39
Fire detector head	Average, industrial	1 470	128	0.09
Flow transmitter, pneumatic	Average, industrial	125	126	1.00
Level indicator, pneumatic	Average, industrial	898	201	0.22
Pressure controller	Average, industrial	40	63	1.58
Pressure indicator, dial, mechanical	Average, industrial	575	178	0.31
Pressure sensor, differential, electronic	Poor, chemical/ship	225	419	1.86
Pressure transmitter	Average, industrial	85 045	806	0.01
Recorder, pen	Average, industrial	26.02	7	0.27
Temperature indicator and alarm	Fair, laboratory	47.2	101	2.14
Temperature indicator, resistance thermometer	Fair, laboratory	212.3	68	0.32
Temperature indicator, bimetal	Average, industrial	165	215	1.30
Temperature trip unit	Average, industrial	120	70	0.58
Thermocouple	Poor, chemical/ship	317	127	0.40
Valve, gate	Poor, chemical/ship	11 564	841	0.07
non-return	Poor, chemical/ship	1 530	101	0.07
solenoid	Poor, chemical/ship	1 804	66	0.04

$$\square \text{ MTBF} = \frac{NT - N_F * \text{MDT}}{N_F} [\text{yr}]$$

$$\square \text{ Mean failure rate } \bar{\lambda} = \frac{1}{\text{MTBF}} [\text{yr}^{-1}]$$



7.1

A batch of 100 identical thermocouples were tested over a 12-week period. Twenty failures were recorded and the corresponding down times in hours were as follows:

5, 6, 7, 8, 4, 7, 8, 10, 5, 4, 8, 5, 4, 5, 6, 5, 4, 9, 8, 6

Calculate:

- (a) Mean down time
- (b) Mean time between failures
- (e) Mean failure rate
- (d) Availability.

7.2

A flow measurement system consists of an orifice plate ($\lambda = 0.75$), differential pressure transmitter ($\lambda = 1.0$), square root extractor ($\lambda = 0.1$) and recorder ($\lambda = 0.1$). Calculate the probability of losing the flow measurement after 0.5 year for the following:

- (a) A single flow measurement system
- (b) Three identical flow measurement systems in parallel
- (c) A system with three orifice plates, three differential pressure transmitters and a middle value selector relay ($\lambda = 0.1$). The selected transmitter output is passed to a single square root extractor and recorder.

Total lifetime operating cost (TLOC)

- ❑ TLOC = initial cost of system (purchase, delivery, installation and commissioning) + cost of failures and maintenance over lifetime of the system + cost of measurement error over lifetime of the system
- ❑ Cost of failures = $[C_R + (C_L + C_P)T_R]\lambda T$, where C_R = average materials cost per repair
 C_L = repair labour cost per hour C_P = process cost per hour, T_R = repair time in hour
- ❑ Cost of maintenance = $(C_M + C_L T_M)m$, C_M cost per service, T_M maintenance time in hours, m is the times for maintenance
- ❑ Total lifetime cost of measurement error = $T \int_{-\infty}^{\infty} C(E)p(E) dE$
- ❑ $TLOC = C_I + [C_R + (C_L + C_P)T_R]\lambda T + (C_M + C_L T_M)m + T \int_{-\infty}^{\infty} C(E)p(E) dE$, C_I initial cost of system

Use the data given in Table Prob. 3 to decide which level measurement system should be purchased. Assume a breakdown maintenance only strategy is practised, each system has the same measurement error, and there is a 10-year total lifetime.

	Parameter	System 1	System 2
Initial cost	£	1000	2000
Materials cost per repair	£	20	15
Labour cost per hour	£	10	10
Process cost per hour	£	100	100
Repair time	h	8	12
Annual failure rate	yr ⁻¹	2.0	1.0

$$\begin{aligned}
 TLOC &= C_I + [C_R + (C_L + C_P)T_R]\lambda T + (C_M + C_L T_M)m + T \int_{-\infty}^{\infty} C(E)p(E) dE \\
 &= 1000 + (20 + (10 + 100) * 8) * 2 * 10 = 19,000 \\
 &\quad 2000 + (15 + (10 + 100) * 12) * 1 * 10 = 15,350
 \end{aligned}$$

- ❑ T_i is the survival time or up time for the i^{th} failure
- ❑ The total up time for N failures is $\sum_{i=1}^N T_i$
- ❑ Mean Time To Failure MTTF = $\frac{\sum_{i=1}^N T_i}{N}$
- ❑ mean failure rate $\lambda = \text{MTTF}$
- ❑ Mean down time = $\frac{\text{total down time}}{\text{number of failures}} = \frac{1}{N_F} \sum_{j=1}^{N_F} T_{Dj}$

